

Physics 230A/146A — Biological Physics

Course syllabus

Course code: 48455 (230A) / 47800 (146A)

Instructor: Jun Allard — jun.allard@uci.edu

Office hours: By request

Meeting time: MWF 10:00–10:50am

Location: Bren Hall 1425

Course website: [Canvas](#)

1 Course description

“Biological physics” means many things to many people. Here, we espouse the view that basic physics — specifically mechanics and elasticity, statistical mechanics (including diffusion, entropy, and free energy), and polymer physics — can help us understand the behavior of biological systems. We will focus on small-scale (cell-scale and molecular-scale) biology, where these areas of physics have had great impact, with an eye towards how cells function in tissues, organs and organisms.

The course is organized as nine lessons, each pairing a physical technique with a specific biological setting drawn where possible from research happening at UC Irvine. Each lesson has an accompanying problem set in which you carry that technique through a real question.

2 Student learning outcomes

Upon successful completion of this course, students should be able to:

- Make order-of-magnitude calculations using sizes, timescales and energy scales to explain biological phenomena.
- Use quantitative models based on physical principles to make predictions of cellular and molecular biological behavior.
- Quantitatively evaluate explanations of cell and molecular biology phenomena in terms of elastic forces, entropy and transport.
- Connect these to biomedical and life sciences research currently happening around the UC Irvine campus.

Above all, the student will have experience going through the full stack of: (1) encountering a biological question or puzzle, usually categorical or conceptual; (2) translating it into a quantitative question; (3) using a model and a mathematical or physics-based technique to analyze that model; and (4) using this analysis to address the original question. Biological physics provides a rich ground of “full stack” questions for this four-step process.

3 Course reading

There is no textbook for the course. Most material is drawn from research articles from the last ten years, and a special effort is made to draw examples from research happening at UC Irvine. There will be around 6 lecture videos ranging from 11min-50min covering material throughout the course.

There are excellent supporting resources, none of them required:

- *Physical Biology of the Cell*, Phillips, Kondev and Theriot, 2nd edition. Most of our topics are covered here, although in a different sequence and with different motivating examples and calculation styles. It was written for a diverse audience, with soft introductions to biology, mathematical methods, and physics.
- *Random Walks in Biology*, Berg.

- *Biological Physics*, Nelson.
- *Mechanics of Motor Proteins and the Cytoskeleton*, Howard.
- *Cell Biology by the Numbers*, Milo and Phillips.

4 Course technology

- **Canvas** for assignment due dates and submission.
- The course website is linked from **Canvas** for assignments and other course material.
- Announcements will be made to the class e-mail list.
- Typesetting software of your choice for preparing solutions — Latex, ipynb, Markdown, Microsoft Word, Mathematica, or anything else that produces a legible PDF. Solutions must be typeset, not handwritten or photographed.

5 Communication

The best way to reach me is by e-mail at jun.allard@uci.edu. Students are responsible for monitoring their UCI e-mail and **Canvas** for course announcements.

6 Assignment details

6.1 Problem sets

There are eight problem sets due throughout the quarter. (There is an additional Problem Set 0 that does not need to be handed in.) Due dates are marked on **Canvas**. **We devote time during scheduled class to working on the problem sets in small groups.**

- Solutions must be typeset using Latex, Markdown, Jupyter, Microsoft Word, Mathematica or another typesetting program, and submitted through **Canvas**.
- Working together is expected and encouraged. The write-up you submit must be your own.
- The lowest score (or missing score) is dropped. No late assignments are accepted.

6.2 Late and incomplete assignments

Late assignments are not accepted. We sometimes go over solutions in class after the due date. The dropped lowest score is the course's built-in accommodation for the ordinary run of things that happen during a quarter, including illness or conference obligations. (You don't need to provide any excuse to drop the lowest or missing score).

6.3 Final week activity

There is no final exam for this course.

7 Course outline

Table 1: Lessons and their problem sets.

Lesson	Topic	Problem set
0	Length, time and energy scales; order-of-magnitude estimation	PS0
1	Mechanics	PS1
2	Higher dimensions	PS2
3	Diffusion	PS3
4	Multi-scale physics	PS4
5	Diffusion and mechanics together	PS5
6	Entropy and entropic forces	PS6
7	Detailed Balance	PS7
8	Polymers	PS8

This plan is subject to change as we proceed through the quarter.

8 Grading

A combination of 80% problem set write-ups (best 7 of 8) and 20% in-person paper quizzes (best 4 of 5).

There will be 5 in-class quizzes. The lowest score (or missed quiz) will be dropped. Quizzes will check understanding of the problem sets.

8.1 AI course policy

Generative AI is allowed when you are out of class, writing up. No generative AI is allowed during small group work. The small group work sessions are your opportunity to gain understanding, and therefore to be able to answer the quiz questions. Quizzes are pen-and-paper, with no electronic assistance of any kind.

9 Campus, Department and course policies

This section summarizes course-specific expectations and points to current public UCI and UC policies and student resources. If details on a linked institutional page change, follow the current institutional policy or procedure. Course-specific notes are marked with a **!** or **i**.

9.1 Classroom etiquette and expectations

i In this course

Much of our scheduled time is spent working in groups. The norm in those sessions is that you will work with each other freely. Mobile devices may be used only for class participation.

Everyone in this course is expected to follow the [Physics & Astronomy Code of Collegial Conduct](#), which describes the department's expectations for respectful, professional, and collaborative conduct.

9.2 Academic integrity

UCI's Academic Integrity Policy applies to all work in this course. See the [Office of Academic Integrity & Student Conduct's Academic Integrity page](#) for current policy, definitions, examples, and procedures. The course-specific rules for generative AI are stated above in the **AI course policy** section.

In this course

Collaboration on the problem sets is expected and encouraged, and much of our class time will be spent in small groups working on these. The write-up that you submit must be your own work. If you are unsure whether a form of collaboration or tool use is permitted, ask the instructor before submitting the work.

9.3 Disability and accessibility accommodations

UCI's [Disability Services Center](#) (DSC) coordinates disability-related accommodations. Students seeking accommodations should follow DSC's current process for obtaining an [Accommodations Letter](#). Approved accommodations will be implemented after the instructor receives the letter from DSC. Please begin the process as early as possible.

9.4 Enrollment changes and withdrawal

You are responsible for your official enrollment in the course. For adding or dropping an individual course or changing enrollment options, see the Registrar's [Add, Drop, and Changes](#) page. Requests after ordinary deadlines are handled through [Enrollment Exceptions](#). Withdrawing from all courses for a term is a separate process; see the Registrar's [Withdrawals](#) page. For department-specific restrictions and procedures, see [Physics & Astronomy Enrollment Policies](#).

9.5 Copyright

Instructor-created lectures and course materials are protected by copyright and UC policy. You may take notes and use course materials provided to you for your own study. You may not otherwise reproduce, distribute, publicly post, or publish course presentations, recordings, or materials without permission except as allowed by law or UC policy. See UC's [Ownership of Course Materials](#) and [Use of Recordings of Course Presentations](#) policies for details. Commercial sale or distribution of course materials may also violate the [UCI Code of Student Conduct](#). You retain copyright in your original work.

9.6 Third-party tools

Some course activities may use third-party technology. UCI's [Learning Technologies](#) page identifies campus-supported tools, and its [Educational Technology review process](#) describes review for accessibility, privacy, and security. Any course-specific instructions for a tool will be provided with the relevant activity.

9.7 Emergency or instructional disruption

Sign up for [zotALERT](#) and review UCI Emergency Management's [Preparedness for Students](#) page. DTEI's [Instructional Continuity](#) page provides campus guidance for disruptions.

In this course

If normal instruction is disrupted, course-specific changes will be communicated through e-mail to the class e-mail list and/or through [Canvas](#).

9.8 Student resources

UCI's [Campus Assist List](#) is a searchable directory for academic support, basic needs, disability services, mental health, safety, housing, and other needs. [Wellness, Health & Counseling Services](#) provides direct access to the Counseling Center, CARE, Disability Services Center, Student Health Center, Campus Social Work, and Student Wellness & Health Promotion.

9.9 Sexual violence, harassment, and nondiscrimination

UCI [CARE](#) provides free and confidential support for students affected by sexual assault, relationship abuse, family violence, or stalking. Faculty and TAs may have reporting obligations when they learn of sexual violence or sexual harassment; see OEOD's [Responsible Employee](#) page for current rules and information about confidential resources. Current UC and UCI discrimination, harassment, and sexual-violence policies are collected on the [Office of Equal Opportunity and Diversity's Policies](#) page.